WORKSHOP AT COLING 2027 · MACAU, CHINA · 9–14 MAY 2027 · FIRST EDITION

# Theories Under Pressure

## Embedding Linguistic Frameworks in Neural Models

LingEmbed 2027 · stergioscha.github.io/LingEmbed-workshop

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### ABOUT

Neural language models are fluent, but they make no explicit commitment to any theory of syntax, semantics or pragmatics. Linguistic frameworks such as Categorial Grammar, HPSG, Minimalism, Construction Grammar, Dynamic Syntax and type-theoretic semantics describe exactly the structure these models leave implicit.

LingEmbed brings together NLP researchers and theoretical linguists around one question: what happens when a linguistic framework is built into a neural model as a working, differentiable component, and can rival theories be compared this way?

You do not need to implement a whole grammar to take part. Complete implementations, partial or phenomenon-specific ones, theory-driven probing, and resources or negative results are all reviewed as full papers.

### TOPICS

Grammars and theories as modules — categorial, Lambek or pregroup grammars compiled into tensor networks or neural parsers; feature structures and unification (HPSG, LFG) as neural components; Minimalist, construction-based or Dynamic Syntax grammars inside neural architectures; type-theoretic semantics (MTT, TTR) as constraints on, or components of, neural models.

Probing for theory — whether pretrained models encode the structure a theory predicts (constituency, dependency, type structure, scope), and how it is distributed.

Theory competition and evaluation — rival frameworks in the same network, compared phenomenon by phenomenon; cognitive and psycholinguistic tests of human-like processing.

Resources and negative results — treebanks, meaning representations, formalisms, grammar implementations, and careful reports of what does not (yet) work.

### TASK-DESIGN SESSION

The workshop closes with an open session to design a future shared task on comparing frameworks inside neural models. Submit a short task-design contribution (desiderata, pilot study, candidate items, feasibility analysis). The session produces a co-authored blueprint; every contributor is credited.

### SUBMISSIONS

Long papers (archival): up to 8 pages
Short papers (archival): up to 4 pages
Position papers: up to 4 pages
Task-design contributions: up to 2 pages
Non-archival: work under review or published elsewhere

Page limits exclude references. One extra camera-ready page; proceedings in the ACL Anthology. ACL templates, double-blind review, direct submission via START (no ARR). Submission site: see the website.

### ORGANISERS

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### PROGRAM COMMITTEE

Lasha Abzianidze · Maxime Amblard · Marianna Apidianaki · Giosuè Baggio · Philippe Blache · Cristiano Chesi · Yan Cong · Robin Cooper · Valeria de Paiva · Yu-yin Hsu · Elisabetta Jezek · Konstantinos Kogkalidis · Alessandro Lenci · Ludovica Pannitto · Sandro Pezzelle · Jakob Prange · He Zhou

### PARTICIPATION

The workshop is hybrid.

### CONTACT

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